「Prove that if $\{a_n\}$ is sequence of integers s.t infinitely many non-zero term, then the convergence bound of $\sum a_n$ is at most 1.

Sol). If $a_n = 1$ $\forall n \in \mathbb{N}$, then $R = 1$.

And, by the Theorem, $R > 1 \iff \limsup_{n \to \infty} \sqrt[n]{|a_n|} < 1$.

Put $\limsup \sqrt[n]{a_n} = \beta < 1$. Then, for $\beta < \alpha < 1$,

$$\exists N \in \mathbb{N} \text{ s.t } n \geq N \Rightarrow \sqrt[n]{|a_n|} < \alpha$$

$$\iff |a_n| < \alpha^n < 1$$

Contradiction.

Section 3.11.

Let $a_n > 0$, $S_n = a_1 + \cdots + a_n$, and $\sum a_n$ diverges. Prove that

a) $\sum \frac{a_n}{1+a_n}$ diverges.

b) $\sum \frac{a_n}{S_n}$ diverges.

c) $\sum \frac{a_n}{S_n^2}$ converges.

d) $\sum \frac{a_n}{1+na_n}$ and $\sum \frac{a_n}{1+n^2 a_n}$ converges?

$a_n \le \frac{1}{2}(1+a_n)$

$\Rightarrow a_n \le 1$

Sol a) Suppose that $\limsup_{n\to\infty} a_n < 1$. Then, there exists $N \in \mathbb{N}$ s.t

$$n \ge N \Rightarrow a_n \le 1$$

$a_n < \alpha + \varepsilon$

Meanwhile, if $a_n \le 1$,

$$\frac{a_n}{2} \le \frac{a_n}{1+a_n}$$

$\alpha - \varepsilon < a_n$

Therefore $\sum \frac{a_n}{1+a_n}$ diverges.

Suppose that $\limsup_{n\to\infty} a_n \ge 1$. Then, there exist infinitely many terms larger than $\frac{1}{2}$.

Since $a_n \ge \frac{1}{2}$ implies

$$a_n \ge \frac{1}{2} \Rightarrow 2a_n \ge 1 \Rightarrow \frac{1}{2a_n + a_n} \le \frac{1}{1+a_n} \Rightarrow \frac{1}{3} \le \frac{a_n}{1+a_n}$$

Therefore diverges.

Alternatively, using contraposition: Suppose $\sum \frac{a_n}{1+a_n}$ converges. Take $N \in \mathbb{N}$ s.t

$$n \ge N \Rightarrow \frac{a_n}{1+a_n} < \frac{1}{2}$$

It equals to $a_n < 1$. Now, for $n \ge N$, $\frac{a_n}{2} \le \frac{a_n}{1+a_n}$ implies $\sum \frac{a_n}{2}$ converges.

(Thanks 현준!)

Sol b). For any $N \in \mathbb{N}$ and $K \ge 0$,

$$\frac{a_{N+1}}{S_{N+1}} + \cdots + \frac{a_{N+K}}{S_{N+K}} \ge \frac{1}{S_{N+K}} \cdot (a_{N+1} + \cdots + a_{N+K}) = \frac{S_{N+K} - S_N}{S_{N+K}} = 1 - \frac{S_N}{S_{N+K}} \quad \cdots (*)$$

If $\sum \frac{a_n}{S_n}$ converges, then there exists $N \in \mathbb{N}$ s.t

$$m \ge n \ge N \Rightarrow \left| \sum_{k=n}^{m} \frac{a_n}{S_n} \right| < \frac{1}{2}.$$

But, since the inequality $(*)$, and $S_n = \sum_{i=0}^{n} a_i$ diverges, we can choose $K \ge N$ s.t

$$\frac{1}{2} \le 1 - \frac{S_N}{S_{N+K}} \le \sum_{i=N+1}^{N+K} \frac{a_i}{S_i}$$

Which is contradiction.

Sol c) Since $S_{n-1} < S_n$ for any $n \in \mathbb{N}$,

$$\frac{a_n}{S_n^2} \le \frac{a_n}{S_{n-1}S_n} = \frac{S_n - S_{n-1}}{S_{n-1} \cdot S_n} = \frac{1}{S_{n-1}} - \frac{1}{S_n} \qquad (n \ge 2)$$

Now, for any $n \ge 2$,

$$\sum_{k=2}^{n} \frac{a_k}{S_k^2} \le \sum_{k=2}^{n} \frac{1}{S_{k-1}} - \frac{1}{S_k} = \frac{1}{S_1} - \frac{1}{S_n} \to \frac{1}{S_1} \quad \text{as } n \to \infty$$

Since S_n diverges to ∞.

Sol d) First,

$$\frac{a_n}{1+n^2 a_n} \le \frac{a_n}{n^2 a_n} = \frac{1}{n^2}, \quad \text{therefore } \sum \frac{a_n}{1+n^2 a_n} \le \sum \frac{1}{n^2}, \text{ converges.}$$

To show the first series diverges,

If $\limsup_{n\to\infty} n a_n < 1$, then there exists $N \in \mathbb{N}$ s.t

$$n \ge N \Rightarrow n a_n < 1$$

which implies that

$$\frac{a_n}{2} \le \frac{a_n}{1+n a_n}$$

therefore $\sum \frac{a_n}{1+n a_n}$ diverges.

If $\limsup_{n\to\infty} n a_n \ge 1$, then there exists infinitely many $n \in \mathbb{N}$ s.t

$$n a_n \ge \frac{1}{2} \qquad 2 n a_n \ge 1$$

Which implies that

$$\frac{a_n}{1+n a_n} \ge \frac{a_n}{2 n a_n + n a_n} \ge \frac{1}{3n}$$

Therefore, $\sum \frac{a_n}{1+n a_n}$ diverges.

$$\frac{a_n}{1+n a_n} \ge \frac{a_n}{n + n a_n} = \frac{a_n}{n(1+a_n)}$$

Bernoui's

If $a > -1$, $(1+a)^n \ge 1 + na$.

Proof. $n = 1$, trivial. $(1+a = 1+a)$.

Suppose $n > 1$, stat. holds if n.

$$\begin{aligned}(1+a)^{n+1} &= (1+a)\cdot(1+a)^n \\ &\ge (1+a)\cdot(1+na) \\ &= (1 + na + a + na^2) \\ &\ge 1 + (n+1)a.\end{aligned}$$

Section 3, 12

Let $a_n > 0$, $\sum a_n$ converges, put $r_n = \sum_{m=n}^{\infty} a_m$

a). Prove $\forall N, K \in \mathbb{N}$,

$$\frac{a_N}{r_N} + \cdots + \frac{a_{N+K}}{r_{N+K}} > 1 - \frac{r_{N+K}}{r_N}$$

Deduce that $\sum \frac{a_n}{r_n}$ diverges.

$$\frac{r_N - a_N}{r_N} = \frac{r_{N+1}}{r_N}$$

b). Prove $\forall n \in \mathbb{N}$,

$$\frac{a_n}{\sqrt{r_n}} < 2(\sqrt{r_n} - \sqrt{r_{n+1}})$$

Deduce that $\sum \frac{a_n}{\sqrt{r_n}}$ converges.

Sol). Since $\sum a_n$ converges, put $A = \sum_{n=1}^{\infty} a_n$. Then,

$r_n \to 0$?

$$A - r_n = \sum_{k=1}^{n-1} a_k$$

And, r_n is decreasing. Now, $r_N > r_{N+i} \Rightarrow \frac{1}{r_N} < \frac{1}{r_{N+i}}$ implies

$$\frac{a_N}{r_N} + \cdots + \frac{a_{N+K}}{r_{N+K}} > \frac{1}{r_N}(a_N + a_{N+1} + \cdots + a_{N+K})$$

$$= \frac{1}{r_N}(r_N - r_{N+K+1})$$

$$= 1 - \frac{r_{N+K+1}}{r_N} > 1 - \frac{r_{N+K}}{r_N}$$

Now, suppose $\sum \frac{a_n}{r_n}$ converges. Then, there exists $N \in \mathbb{N}$ s.t for all $K \in \mathbb{N}$,

$$\left| \sum_{n=N}^{N+K} \frac{a_n}{r_n} \right| < \frac{1}{2}.$$

Meanwhile, $\lim_{n\to\infty} r_n = 0$ because for any $\varepsilon > 0$, we can choose $N \in \mathbb{N}$ s.t

$$n \geq N \Rightarrow |r_n| = r_n = \sum_{m=n}^{\infty} r_m = A - \sum_{k=1}^{n-1} a_k < \varepsilon$$

Therefore, we can choose $R \in \mathbb{N}$ for fixed N,

$$K \geq R \Rightarrow \left| \sum_{n=N}^{N+K} \frac{a_n}{r_n} \right| > 1 - \frac{r_{N+K}}{r_N} \geq \frac{1}{2}.$$

It is contradiction.

Sol b). Since $a_n = r_n - r_{n+1} = (\sqrt{r_n} - \sqrt{r_{n+1}})(\sqrt{r_n} + \sqrt{r_{n+1}})$ $(n \geq 1)$

$$\frac{a_n}{\sqrt{r_n}} = (\sqrt{r_n} - \sqrt{r_{n+1}})\left(1 + \frac{\sqrt{r_{n+1}}}{\sqrt{r_n}}\right)$$

$$< (\sqrt{r_n} - \sqrt{r_{n+1}})(1 + 1) = 2(\sqrt{r_n} - \sqrt{r_{n+1}})$$

Therefore,

$$\sum_{k=1}^{n} \frac{a_k}{\sqrt{r_k}} < \sum_{k=1}^{n} 2(\sqrt{r_k} - \sqrt{r_{k+1}})$$

$$= 2(\sqrt{r_1} - \sqrt{r_{n+1}}) \to 2\sqrt{r_1} \quad \text{as } n \to \infty.$$

Thus $\sum \frac{a_n}{\sqrt{r_n}}$ converges.

「Section 3.13

Given absol. convergent $\sum a_n$, $\sum b_n$, $\sum_{r=n}^{\infty} \sum_{k=1}^{n} a_{n+k} b_k$ converges absolutely.

Proof. Denote $A_n = \sum_{k=0}^{n} |a_k|$, $B_n = \sum_{k=0}^{n} |b_k|$, and $C_n = \sum_{k=0}^{n} \left| \sum_{r=0}^{k} a_{k-r} \cdot b_r \right|$ $\beta_n = B - B_n$

Put $A = \lim_{n\to\infty} A_n$, $B = \lim_{n\to\infty} B_n$

$$C_n = |a_0 b_0| + |a_1 b_0 + a_0 b_1| + |a_2 b_0 + a_1 b_1 + a_0 b_2| + \cdots + |a_n b_0 + \cdots + a_0 b_n|$$

$$\leq |a_0| B_n + |a_1| B_{n-1} + \cdots + |a_n| B_0$$

$$= |a_0|(B - \beta_n) + |a_1|(B - \beta_{n-1}) + \cdots + |a_n|(B - \beta_0)$$

$$= A_n B - (|a_0| \beta_n + |a_1| \beta_{n-1} + \cdots + |a_n| \beta_0)$$

$$\leq A_n B \to AB$$

Therefore, C_n is monotone increasing bounded sequence, it is converges.

Section 3.14.

Let $\{s_n\}$ be a complex numbers sequence, and define

$$\sigma_m = \frac{s_1 + s_2 + \cdots + s_m}{m} \quad (m \in \mathbb{N})$$

$$(m\sigma_m - n\sigma_n)$$

Q. If $\lim_{n\to\infty} s_n = s$, then $\lim_{n\to\infty} \sigma_n = s$

Sol) Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t

$$n \geq N \Rightarrow |s_n - s| < \frac{\varepsilon}{2}$$

meanwhile, observe that for $m > N$,

$$|m\sigma_m - ms| = |(s_1 - s) + \cdots + (s_N - s) + (s_{N+1} - s) + \cdots + (s_m - s)|$$

$$\leq |(s_1 - s) + \cdots + (s_N - s)| + |s_{N+1} - s| + \cdots + |s_m - s|$$

$$\leq |(s_1 - s) + \cdots + (s_N - s)| + m \cdot \frac{\varepsilon}{2} \quad \left(m\frac{\varepsilon}{2} \text{ can be replaced by } (m-N+1)\frac{\varepsilon}{2}\right)$$

Taking $\frac{1}{m}$ both side, then

$$|\sigma_m - s| \leq \frac{1}{m} \cdot |(s_1 - s) + \cdots + (s_N - s)| + \frac{\varepsilon}{2}$$

Now, Take $M \in \mathbb{N}$ s.t $|(s_1 - s) + \cdots + (s_N - s)| < M\frac{\varepsilon}{2}$, then

$$m \geq M \Rightarrow |\sigma_m - s| < \varepsilon.$$

a-1) If $\{s_n\}$ is a real numbers, then

$$\liminf_{n\to\infty} s_n \leq \liminf_{n\to\infty} \sigma_n \leq \limsup_{n\to\infty} \sigma_n \leq \limsup_{n\to\infty} s_n$$

Proof. Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t

$$n \geq N \Rightarrow s_n < \limsup_{n\to\infty} s_n + \frac{\varepsilon}{2}$$

$$\sigma_n = \frac{s_1 + \cdots + s_N + s_{N+1} + \cdots + s_n}{n} = \frac{1}{n}(s_1 + \cdots + s_N) + \frac{1}{n}(s_{N+1} + \cdots + s_n)$$

$$\leq \frac{1}{n}(s_1 + \cdots + s_N) + \frac{1}{n} \cdot (n - N) \cdot \left(\limsup_{n\to\infty} s_n + \frac{\varepsilon}{2}\right)$$

$$\leq \frac{1}{n}(s_1 + \cdots + s_N) + \limsup s_n + \frac{\varepsilon}{2}$$

Take $M \in \mathbb{N}$ s.t $(s_1 + \cdots + s_N) < M \cdot \frac{\varepsilon}{2}$. Then, for $n \geq M$,

$$\sigma_n \leq \limsup_{n\to\infty} s_n + \varepsilon.$$

Since $\varepsilon > 0$ was chosen arbitrarily, $\limsup_{n\to\infty} \sigma_n \leq \limsup_{n\to\infty} s_n$.

d). Put $a_n = s_{n+1} - s_n$; $n \geq 1$. Show that

$$s_{n+1} - \sigma_n = \frac{1}{n} \cdot \sum_{k=1}^{n} k a_k$$

And deduce that $\lim_{n\to\infty} n a_n = 0$ and $\lim_{n\to\infty} \sigma_n = s$ implies $\lim_{n\to\infty} s_n = s$.

Proof. Since

$$\begin{aligned}
\sum_{k=1}^{n} k a_k &= a_1 + 2a_2 + 3a_3 + \cdots + n a_n \\
&= (s_2 - s_1) + 2(s_3 - s_2) + \cdots + n(s_{n+1} - s_n) \\
&= n s_{n+1} - s_n - s_{n-1} - \cdots - s_1 \\
&= n s_{n+1} - (s_1 + \cdots + s_n) \\
&= n s_{n+1} - n \cdot \sigma_n
\end{aligned}$$

Therefore,

$$\lim_{n\to\infty} s_n = \lim_{n\to\infty} \Big(\sigma_n + \underbrace{\frac{1}{n} \cdot \sum_{k=1}^{n} k a_k}_{=0,\ \text{by a)}} \Big) = \lim_{n\to\infty} \sigma_n = s.$$

e) Assume $\{n a_n\}$ bounded and $\lim_{n\to\infty} \sigma_n = s$, then $\lim_{n\to\infty} s_n = s$.

Proof. Let $M > 0$ be a bound of $\{n a_n\}$.

Observe that for $n > m$,

$$\begin{aligned}
s_n - \sigma_n &= s_n - \frac{1}{n} \cdot (s_1 + \cdots + s_n) \\
&= s_n - \frac{m}{n-m} \left(\frac{1}{m} - \frac{1}{n} \right) (s_1 + \cdots + s_n) \\
&= s_n - \frac{m}{n-m} \left(\frac{1}{m} (s_1 + \cdots + s_m + s_{m+1} + \cdots + s_n) - \frac{1}{n} (s_1 + \cdots + s_n) \right) \\
&= s_n - \frac{m}{n-m} \left(\frac{1}{m} \cdot [m \sigma_m + (s_{m+1} + \cdots + s_n)] - \sigma_n \right) \\
&= s_n - \frac{m}{n-m} \left(\sigma_m - \sigma_n + \frac{1}{m} \cdot [s_{m+1} + \cdots + s_n] \right) \\
&= \frac{m}{n-m} (\sigma_n - \sigma_m) + \frac{1}{n-m} \cdot \sum_{k=m+1}^{n} (s_n - s_k)
\end{aligned}$$

($n > m$, then $\frac{1}{n} = \frac{m}{n-m} \left(\frac{1}{m} - \frac{1}{n} \right)$, it is derived by $\frac{1}{AB} = \frac{1}{B-A} \left(\frac{1}{A} - \frac{1}{B} \right)$)

$-1 + m\varepsilon \leq n \iff \frac{m}{n-m+1} \leq \frac{1}{\varepsilon}$

$n - \varepsilon < m + m\varepsilon \iff \frac{n-m}{\ } < \varepsilon$

Meanwhile, since $|n a_n| = n|s_{n+1} - s_n| < M$, for $i \geq m+1$,

$$\begin{aligned}
|s_n - s_i| &= |s_n - s_{n-1} + s_{n-1} - s_{n-2} + s_{n-2} - \cdots + s_{i+1} - s_i| \\
&\leq |s_n - s_{n-1}| + |s_{n-1} - s_{n-2}| + \cdots + |s_{i+1} - s_i| \\
&\leq \frac{M}{n-1} + \frac{M}{n-2} + \cdots + \frac{M}{i} \leq \frac{n-i}{i} \cdot M \leq \frac{n-i}{m+1} \cdot M \leq \frac{n-m-1}{m+1} \cdot M.
\end{aligned}$$

Given $\varepsilon > 0$ and for each $n \in \mathbb{N}$, take $m \in \mathbb{N}$ satisfying $m \overset{(1)}{\leq} \frac{n-\varepsilon}{1+\varepsilon} \overset{(2)}{<} m+1$

Then, $m < n$ and $(1) \iff m + m\varepsilon + \varepsilon \leq n \iff \frac{m+1}{n-m} \leq \frac{1}{\varepsilon}$

and $(2) \iff n - \varepsilon \leq m+1 + \varepsilon M + \varepsilon \iff \frac{n-m-1}{m+2} < \varepsilon$ implies if $i \geq m+1$, $|s_n - s_i| \leq \frac{n-m-1}{m+1} \cdot M < \varepsilon M$.

Finally

$$\begin{aligned}
\limsup_{n\to\infty} |s_n - s| &\leq \limsup_{n\to\infty} |s_n - \sigma_n| + \limsup_{n\to\infty} |\sigma_n - s| = \limsup_{n\to\infty} |s_n - \sigma_n| \\
&\leq \limsup_{n\to\infty} \left(\frac{m}{n-m} |\sigma_n - \sigma_m| \right) + \limsup_{n\to\infty} \left(\frac{1}{n-m} \cdot \sum_{k=m+1}^{n} |s_n - s_k| \right) \\
&\leq \frac{1}{\varepsilon} \limsup_{n\to\infty} |\sigma_n - \sigma_m| + \varepsilon M = \varepsilon M
\end{aligned}$$

Since $\varepsilon > 0$ was arbitrary, $\lim_{n\to\infty} s_n = s$.

Section 3.16.

Fix $\alpha > 0$, and choose $x_1 > \sqrt{\alpha}$. Define $\{x_n\}$ as recursively:

$$x_{n+1} := \frac{1}{2} \cdot \left(x_n + \frac{\alpha}{x_n} \right)$$

a). $\{x_n\}$ decreases monotonically, and $\lim_{n \to \infty} x_n = \sqrt{\alpha}$

Proof. observe that

$$x_{n+1} \leq x_n \iff \alpha \leq x_n^2 \iff \frac{\alpha}{x_n} \leq x_n$$

For $n=1$, it is trivially true. Suppose for $n \in \mathbb{N}$, it is true. Then,

$$x_{n+1} = \frac{1}{2} \cdot \left(x_n + \frac{\alpha}{x_n} \right) \leq \frac{1}{2} \cdot \left(x_n + x_n \right) = x_n.$$

Point: $\frac{\alpha}{x_n} \leq x_n$

By induction, $\{x_n\}$ is monotone decreasing. And $x_n \geq \sqrt{\alpha}$ for all $n \in \mathbb{N}$,

$\{x_n\}$ is decreasing bounded, therefore converges. Put $x = \lim_{n \to \infty} x_n \geq \sqrt{\alpha}$. Now,

$$x = \lim_{n \to \infty} x_{n+1} = \frac{1}{2} \left(\lim_{n \to \infty} x_n + \frac{\alpha}{\lim_{n \to \infty} x_n} \right) = \frac{1}{2} \left(x + \frac{\alpha}{x} \right) \Rightarrow x = \sqrt{\alpha}.$$

b) Put $\varepsilon_n = x_n - \sqrt{\alpha}$ (That is, error between x_n and $\sqrt{\alpha}$, magnitude for approximation)

Show that $\varepsilon_{n+1} = \frac{\varepsilon_n^2}{2x_n} < \frac{\varepsilon_n^2}{2\sqrt{\alpha}}$ and that $\varepsilon_{n+1} < 2\sqrt{\alpha} \cdot \left(\frac{\varepsilon_1}{2\sqrt{\alpha}} \right)^{2^n}$ for $n \in \mathbb{N}$

Proof Since

$$\varepsilon_{n+1} = x_{n+1} - \sqrt{\alpha} = \frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right) - \sqrt{\alpha} = \frac{1}{2} \cdot \frac{\varepsilon_n^2 + 2\varepsilon_n \sqrt{\alpha} + 2\alpha}{x_n} - \sqrt{\alpha} = \frac{\varepsilon_n^2}{2x_n} + \frac{\sqrt{\alpha}}{x_n} \cdot (\varepsilon_n + \sqrt{\alpha}) - \sqrt{\alpha} = \frac{\varepsilon_n^2}{2x_n}$$

and $x_n \geq \sqrt{\alpha}$ gives $\varepsilon_{n+1} < \frac{\varepsilon_n^2}{2\sqrt{\alpha}}$. Now, inductively,

$$\varepsilon_{n+1} < \frac{\varepsilon_n^2}{2\sqrt{\alpha}} < \frac{1}{2\sqrt{\alpha}} \cdot \left(\frac{\varepsilon_{n-1}^4}{4\alpha} \right) = \frac{1}{2\sqrt{\alpha}} \cdot \frac{1}{(2\sqrt{\alpha})^2} \cdot \varepsilon_{n-1}^4 < \cdots \quad \left(\sum_{k=0}^{n-1} 2^k = 2^n - 1 \right)$$

$$< \left(\frac{1}{2\sqrt{\alpha}} \right)^{1+2+2^2+\cdots+2^{n-1}} \cdot \varepsilon_{n-(n-1)}^{2^n} = 2\sqrt{\alpha} \cdot \left(\frac{\varepsilon_1}{2\sqrt{\alpha}} \right)^{2^n}$$